



National
Qualifications
2024

2024 Human Biology

Higher - Paper 1

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question	Response	Mark
1.	A	1
2.	C	1
3.	D	1
4.	C	1
5.	C	1
6.	A	1
7.	B	1
8.	D	1
9.	A	1
10.	D	1
11.	D	1
12.	D	1
13.	C	1
14.	B	1
15.	C	1
16.	B	1
17.	C	1
18.	A	1
19.	A	1
20.	C	1
21.	A	1
22.	B	1
23.	B	1
24.	B	1
25.	D	1

[END OF MARKING INSTRUCTIONS]



National
Qualifications
2024

2024 Human Biology

Higher - Paper 2

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)		A T A G C A	1	
	(b)	(i)	Any value from 92 - 98	1	Accept any range between 92 - 98
		(ii)	1. allows nucleotides to be added. OR allows DNA to be amplified/amplification to start. OR provides a starting point for amplification. (1) 2. adds/joins (complementary) nucleotides (to a DNA strand/primer). OR amplifies DNA. (1)	2	Accept binds to complementary/target/specific sequence/(section of) DNA. Do not accept bases for nucleotides.
	(c)		1. Solving crimes. 2. Identifying people. 3. Diagnose genetic disorders/mutations. Any 2 from 3	2	
2.	(a)	(i)	100	1	
		(ii)	39/39.3/39.29	1	
	(b)	(i)	2679	1	
		(ii)	To allow a valid comparison between countries. OR (As the countries) have different population sizes/numbers.	1	Accept a comparison, e.g. England has a greater/larger population than the other countries.

Question			Expected response	Max mark	Additional guidance												
3.	(b)		Axes have correct scale and labels. (1) Points correctly plotted and line drawn. (1) <table><tr><th>Concentration of copper sulfate (mmol/l)</th><th>Volume of oxygen collected (cm³)</th></tr><tr><td>0</td><td>82</td></tr><tr><td>5</td><td>68</td></tr><tr><td>10</td><td>45</td></tr><tr><td>25</td><td>25</td></tr><tr><td>50</td><td>16</td></tr></table>	Concentration of copper sulfate (mmol/l)	Volume of oxygen collected (cm ³)	0	82	5	68	10	45	25	25	50	16	2	Candidates cannot access the scale mark if the plotted points use less than half the graph paper. If candidate draws a bar graph they cannot access the scale mark.
Concentration of copper sulfate (mmol/l)	Volume of oxygen collected (cm ³)																
0	82																
5	68																
10	45																
25	25																
50	16																
	(c)		As the inhibitor/copper sulfate concentration increases, the catalase activity decreases.	1													

Question			Expected response	Max mark	Additional guidance
4.	(a)	(i)	Y and Z	1	
		(ii)	Pyruvate	1	
		(iii)	Citrate	1	
	(b)		Inner membrane (in mitochondria from muscle cells) has a greater surface area/more folds/is larger/longer. (1) For electron transport chains/ATP synthase/hydrogen ions to be pumped across. (1)	2	
	(c)		They have many mitochondria. OR They have a large blood supply. OR They have a high concentration of myoglobin. OR Their major storage fuel is fat.	1	

Question			Expected response	Max mark	Additional guidance
5.	(a)	(i)	(FSH) stimulates development/maturation/growth of the follicle (in the ovary). (1) The (developing) follicle releases oestrogen. (1)	2	
		(ii)	Oestrogen stimulates proliferation/repair/thickening/regeneration of the endometrium. OR Oestrogen prepares the endometrium for implantation.	1	
		(iii)	It stimulates ovulation/release of an egg/ovum (from the follicle/ovary).	1	
	(b)	(i)	14	1	
		(ii)	Corpus luteum	1	
	(c)	(i)	Progesterone (concentration) decreases (after day 25/towards the end of the cycle). OR Progesterone (concentration) does not remain high (after day 25/towards the end of the cycle).	1	
		(ii)	Treatment - IVF. (1) Description - eggs are mixed with sperm/fertilised in a culture dish/outside the body. OR Description - eggs are (surgically) removed and mixed with sperm/fertilised. (1)	2	

Question			Expected response	Max mark	Additional guidance									
6.	(a)		Could lead to a false positive/negative. OR Unnecessary (diagnostic) tests could be carried out.	1										
	(b)		(Amniocentesis has) a lower risk/chance of miscarriage. OR CVS has a higher risk/chance of miscarriage.	1	Accept amniocentesis is safer for the fetus. OR CVS is more dangerous for the fetus. OR Amniocentesis may be carried out later in the pregnancy. OR It was too late in the pregnancy for CVS.									
	(c)	(i)	Karyotype	1										
		(ii)	Autosomes	1	Accept autosomal									
7.	(a)	(i)	300	1										
		(ii)	2 : 48 : 3	1										
		(iii)	<table><tr><th>Part of body</th><th>Change in blood flow rate</th><th>Process controlling blood flow</th></tr><tr><td>Intestines</td><td>decrease</td><td>vasoconstriction</td></tr><tr><td>Skeletal muscles</td><td>increase</td><td>vasodilation</td></tr></table> 1 mark for part of body. 1 mark for process.	Part of body	Change in blood flow rate	Process controlling blood flow	Intestines	decrease	vasoconstriction	Skeletal muscles	increase	vasodilation	2	Accept description of smooth muscle in artery contracting/relaxing for vasoconstriction/vasodilation.
Part of body	Change in blood flow rate	Process controlling blood flow												
Intestines	decrease	vasoconstriction												
Skeletal muscles	increase	vasodilation												
	(b)	(i)	150	1										
		(ii)	A sympathetic nerve/neuron releases/secretates noradrenaline	1										

Question			Expected response	Max mark	Additional guidance
8.	(a)	(i)	Plasma	1	
		(ii)	Pressure filtration	1	
	(b)		Blood is at a higher pressure (in A). OR Vasoconstriction/vasodilation occurs in A but not in D.	1	Accept artery/vein for A/D.
	(c)	(i)	Walls are thin/only one cell thick.	1	Accept they provide a large surface area (for diffusion). Accept endothelial/endothelium/epithelial/epithelium for walls.
		(ii)	(Plasma) Protein	1	
	(d)	(i)	Absorbs excess tissue fluid. OR Carries lymph back to the circulatory system.	1	
		(ii)	They prevent the backflow of fluid/lymph. OR They make sure that the fluid/lymph only flows in one direction.	1	

Question			Expected response	Max mark	Additional guidance
9.	(a)		(An inflatable) cuff stops/restricts blood flow/circulation (in the artery). OR (An inflatable) cuff deflates (gradually) so blood flow/circulation (in artery) can resume. (1) (Systolic blood) pressure is measured when blood starts to flow/a pulse is detected. (1)		
	(b)	(i)	Blood pressure	1	
		(ii)	(Consuming) energy drinks increases (diastolic) blood pressure. OR (Consuming) energy drinks has no effect on systolic blood pressure.		
		(iii)	97	1	
	(c)		Hypertension	1	

Question			Expected response	Max mark	Additional guidance
10.	(a)	(i)	Age/gender/ethnicity/other health conditions	1	
		(ii)	Type 1 diabetes may cause low vitamin D (concentrations). OR Type 1 diabetes may have occurred before/earlier/first. OR The results are only from one study/small sample size. OR More than half/59% of the control group have low vitamin D concentrations (but not type 1 diabetes).	1	
		(iii)	118	1	
		(iv)	Repeat the study with larger groups/more individuals in each group. OR Repeat the study and calculate an average.	1	
	(b)	(i)	From January to May the average hours of sunlight increases/goes from 36 to 184/increases by 148 hours from 36. (1) From May to December the average hours of sunlight decreases/goes from 184 to 32 hours/decreases by 152 hours from 184. (1)	2	Award 1 mark if candidates state it increases to May and then it decreases if no other marks are awarded. Answer must indicate hours for at least one figure.
		(ii)	October to March	1	Accept January to March and October to December.

Question			Expected response	Max mark	Additional guidance
11.	(a)		Displacement/decay	1	
	(b)		Working memory (model)	1	
	(c)		STM/it has a limited capacity. OR 11 numbers/it exceeds capacity of STM.	1	Accept can only hold 5 to 9 numbers/items. Accept can only hold 7 ± 2 numbers/items. Accept any value in the range 5 to 9.
	(d)	(i)	(They could be allocated) by a computer (program)/by selecting names out of a hat.	1	Do not accept randomisation only. Accept any other suitable example of random allocation.
		(ii)	(Group 1 used) organisation which allows words to be transferred/encoded (from STM) to LTM. OR Group 2 did not use organisation so fewer words were transferred/encoded (from STM) to LTM.	1	
	(e)		Contextual cues	1	

Question			Expected response	Max mark	Additional guidance
12.	(a)		Synapse/synaptic cleft	1	
	(b)	(i)	Diverging	1	
		(ii)	More than one neuron/muscle/finger is affected/stimulated at the same time. OR Impulse/neurotransmitter is sent to many neurons/muscles/fingers giving coordinated movement.	1	
	(c)	(i)	Myelin (sheath)	1	
		(ii)	It increases the speed of impulses. OR It insulates the axon.	1	
	(d)	(i)	Fewer receptors/no receptors for the neurotransmitter to bind to. OR Fewer receptors/no receptors so impulses/neurotransmitters cannot cause muscle contraction. OR The threshold (initiating muscle contraction) cannot be reached.	1	Accept neurotransmitters cannot bind to the receptors.
		(ii)	T lymphocytes respond to self-antigens/antigens on own (healthy) cells. (1) Attacking/destroying/killing the body's own (healthy) cells. (1)	2	Accept T lymphocytes fail to recognise self-antigens/antigen on own (healthy) cells. Accept apoptosis of the body's own cells.
		(iii)	10 320	1	

Question			Expected response	Max mark	Additional guidance
13.	(a)		1. B lymphocytes have receptors which recognise antigens on pathogen/foreign antigens. 2. (Following recognition) they form clones/a clonal population/copies of themselves. OR They undergo clonal selection. 3. B lymphocytes produce antibodies. 4. Antibodies bind to antigens on the pathogen. OR The antigen-antibody complex can be destroyed/removed by phagocytosis. Any 3 from 4	3	
	(b)		Allergy/allergic reaction/hypersensitive response	1	

Question			Expected response	Max mark	Additional guidance
14.	(a)		DNA/RNA/nucleic acid/chromosome	1	
	(b)		(Inactivated) pathogen toxins/dead pathogens/weakened pathogens	1	
	(c)	(i)	To establish their safety/effectiveness.	1	
		(ii)	Neither the subjects nor the researchers know which group the subjects are in. OR Neither the subject nor the researchers know who has been given the placebo/vaccine.	1	
	(d)		Antigenic variation occurs. (1) Memory cells/memory lymphocytes are not effective against/do not recognise the changed/mutated/different/new antigen. OR Memory cells/memory lymphocytes are not effective against/do not recognise the changed/mutated/alterd pathogen/virus. (1)	2	

Question			Expected response	Max mark	Additional guidance
15.	A		1. A somatic cell is any cell (in the body) other than cells involved in reproduction/germline cells. 2. Germline cells are gametes/ sperm and egg/ova and (stem) cells that divide/differentiate to form/produce gametes. 3. Somatic cells divide by mitosis to form (more) somatic cells. 4. Mitosis of germline cells produces (more) germline cells. 5. Meiosis of germline cell produces gametes. 6. Somatic cells/germline stem cells are diploid and gametes are haploid. 7. Meiosis has two stages/divisions. 8. First stage/division separates homologous/matching/pairs of chromosomes. 9. Second stage/division separates chromatids. 10. Somatic cells/germline cells divide by mitosis to maintain the (diploid) chromosome number/ complement. 11. Diploid cells have 23 pairs of homologous/matching chromosomes. OR Haploid cells/gametes contain 23 single chromosomes/one set of 23 chromosomes. 8 marks from 11	8	3. Accept identical daughter cells for somatic cells. a. Award mark for germline cells can divide by mitosis and meiosis if candidate has not been awarded both points 4 and 5.

Question			Expected response	Max mark	Additional guidance
15.	B		1. mRNA/primary transcript is transcribed/produced from DNA in the nucleus. 2. RNA polymerase unwinds the DNA and breaks the (hydrogen) bonds between the bases/strands. OR RNA polymerase unwinds the DNA and unzips the strands. 3. RNA polymerase adds complementary RNA nucleotides. OR RNA polymerase allows complementary base pairing to occur. 4. RNA contains the (base) uracil instead of thymine. OR Uracil in RNA is complementary to/pairs with adenine. 5. The primary mRNA/transcript undergoes (RNA) splicing. 6. Primary mRNA/transcript has introns and exons. 7. Introns are non-coding regions, exons are coding. 8. Introns are removed and exons are joined/retained to form mature mRNA/transcript. 9. Different mature mRNA/mature transcripts/proteins can be produced as a result of alternative (RNA) splicing. 10. Different mature mRNA/mature transcripts/proteins can be produced depending on which exons are retained. 11. The order of exons is not changed. 8 marks from 11	8	

General marking principles for Higher Human Biology

Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

- (a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- (b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.
- (c) Do not award half marks.
- (d) Where a candidate makes an error in the first part of a question, award marks for subsequent answers that are correct with regard to this original error. Do not penalise candidates more than once for the same error.
- (e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final answer (including units, if appropriate) on its own.
- (f) Candidates should not use bulleted lists to answer extended-response questions. They must respond to the 'command' word as appropriate and provide extended answers to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.
- (g) In the detailed marking instructions, if a word is **underlined** then it is essential; if a word is **(bracketed)** then it is not essential.
- (h) In the detailed marking instructions, words separated by / are **alternatives**.
- (i) A correct response can be negated if the candidate includes:
 - an extra, incorrect, response
 - additional information that contradicts the correct response
- (j) Where the candidate is instructed to choose one question to answer but instead answers two questions, mark both responses and award the higher mark.
- (k) Unless otherwise required by the question, the use of abbreviations (for example DNA, ATP) or chemical formulae (for example CO₂, H₂O) are acceptable alternatives to naming.
- (l) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, do not penalise candidates repeatedly.
- (m) If incorrect spelling is given:
 - If the correct word is recognisable then award the mark.
 - If the word can easily be confused with another biological term then **do not** award the mark, for example glucagon and glycogen.

(n) Presentation of data:

- If a candidate provides two graphs, in response to one question, mark both and award the higher mark.
- If a question asks for a particular type of graph/chart and the candidate gives the wrong type, do not award full marks. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label. If the x and y data are transposed, then do not award the scale and label mark.
- If the graph uses less than 50% of the axes then do not award the scale and label mark.
- If 0 is plotted when no data for this is given, then do not award the plot mark – candidates should only plot the data given.

(o) Only award marks for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give or state**, they need only answer or present in brief form
- **describe**, they must provide a statement as opposed to simply one word
- **explain**, they must provide a reason for the information given
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined
- **calculate**, they must determine a number from given facts, figures or information
- **predict**, they must indicate what may happen based on available information
- **suggest**, they must apply their knowledge and understanding to a new situation

2024 Higher Human Biology Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	A	78	13	3	6	0	0
2	C	7	13	77	3	0	0
3	D	11	2	7	80	0	0
4	C	12	14	54	20	0	0
5	C	7	5	84	4	0	0
6	A	75	5	16	4	0	0
7	B	9	76	8	7	0	0
8	D	1	4	1	94	0	0
9	A	52	14	24	10	0	0
10	D	10	25	14	50	0	0
11	D	3	13	14	70	0	0
12	D	20	8	39	33	0	0
13	C	13	5	75	7	0	0
14	B	2	64	6	28	0	0
15	C	27	13	42	17	0	0
16	B	6	56	33	5	0	0
17	C	11	12	60	17	0	0
18	A	82	9	4	5	0	0
19	A	87	5	4	4	0	0
20	C	9	4	72	15	0	0
21	A	33	6	44	16	0	0
22	B	13	78	5	4	0	0
23	B	8	56	24	12	0	0
24	B	15	61	12	12	0	0
25	D	11	6	7	76	0	0

2024 Higher Human Biology Section 2 Marks Data Analysis

2024 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1a	1	0.9	93.8%	17	0.3%
Q1bi	1	0.7	66.8%	40	0.3%
Q1bii	2	0.9	44.0%	58	3.0%
Q1c	2	1.4	70.3%	52	3.9%
Q2ai	1	0.8	84.3%	24	0.1%
Q2aii	1	0.3	34.1%	40	1.1%
Q2bi	1	0.7	73.3%	31	2.0%
Q2bii	1	0.1	14.9%	32	5.3%
Q3ai	2	1.3	67.2%	36	0.3%
Q3aii	1	0.1	6.6%	11	4.7%
Q3aiii	1	0.3	31.3%	26	5.5%
Q3b	2	1.6	80.1%	34	0.8%
Q3c	1	0.3	27.4%	50	1.2%
Q4ai	1	0.4	41.2%	49	0.4%
Q4aii	1	0.6	63.4%	50	3.6%
Q4aiii	1	0.6	64.4%	60	6.4%
Q4b	2	0.7	33.0%	59	4.5%
Q4c	1	0.7	65.6%	50	5.4%
Q5ai	2	0.8	42.2%	66	8.1%
Q5aii	1	0.5	48.1%	51	5.3%
Q5aiii	1	0.6	62.7%	47	7.0%
Q5bi	1	0.7	65.9%	36	3.5%
Q5bii	1	0.7	65.2%	59	4.3%
Q5ci	1	0.6	60.7%	49	2.5%
Q5cii	2	1.2	61.0%	40	4.2%
Q6a	1	0.5	49.2%	45	4.5%
Q6b	1	0.6	60.5%	48	7.1%
Q6ci	1	0.6	56.2%	59	13.6%
Q6cii	1	0.3	32.6%	55	8.8%
Q7ai	1	0.5	51.8%	45	1.2%
Q7aii	1	0.6	64.3%	32	1.7%
Q7aiii	2	1.1	53.2%	50	5.4%
Q7bi	1	0.5	49.8%	49	8.2%
Q7bii	1	0.3	29.8%	55	9.3%
Q8ai	1	0.6	59.2%	29	4.0%
Q8aii	1	0.5	50.0%	64	16.8%
Q8b	1	0.4	38.7%	32	3.2%
Q8ci	1	0.5	46.4%	37	7.1%
Q8cii	1	0.5	52.0%	54	6.7%
Q8di	1	0.3	29.8%	56	14.6%
Q8dii	1	0.2	20.6%	36	5.0%
Q9a	2	0.5	23.7%	47	12.1%
Q9bi	1	0.3	32.9%	36	2.4%
Q9bii	1	0.5	49.8%	28	1.3%
Q9biii	1	0.5	51.0%	50	7.7%
Q9c	1	0.4	39.1%	52	7.6%
Q10ai	1	0.6	57.2%	22	2.3%
Q10aii	1	0.2	19.1%	23	5.7%
Q10aiii	1	0.5	47.6%	36	3.5%
Q10aiv	1	0.4	42.7%	30	1.6%
Q10bi	2	0.8	40.8%	50	1.6%
Q10bii	1	0.8	81.0%	26	1.1%
Q11a	1	0.7	67.3%	48	5.4%
Q11b	1	0.3	26.3%	47	22.8%
Q11c	1	0.6	63.6%	34	1.4%
Q11di	1	0.5	48.7%	33	8.1%
Q11dii	1	0.2	21.8%	40	1.6%
Q11e	1	0.5	47.0%	44	7.0%
Q12a	1	0.3	34.7%	26	2.9%
Q12bi	1	0.7	73.9%	46	5.5%
Q12bii	1	0.1	11.0%	31	11.4%
Q12ci	1	0.8	78.4%	48	5.0%
Q12cii	1	0.5	46.8%	53	8.4%
Q12di	1	0.1	10.8%	29	6.2%
Q12dii	2	0.5	23.5%	54	13.9%
Q12diii	1	0.5	52.7%	35	4.9%
Q13a	3	0.9	28.6%	68	10.6%
Q13b	1	0.6	60.5%	57	10.1%
Q14a	1	0.3	26.7%	27	5.4%
Q14b	1	0.3	34.6%	45	15.3%
Q14ci	1	0.8	85.0%	27	2.4%
Q14cii	1	0.6	58.1%	45	4.9%
Q14d	2	0.6	27.5%	63	4.6%
Q15A	8	2.8	35.0%	60	4.3%
Q15B	8	4.1	51.5%	72	0.0%
Paper	95	44.7	47.1%		

2024 Higher Human Biology Section 2 Distribution of Marks Analysis

2024 Item	0	1	2	3	4	5	6	7	8
Q1a	6	94							
Q1bi	33	67							
Q1bii	40	31	28						
Q1c	18	22	59						
Q2ai	16	84							
Q2aii	66	34							
Q2bi	27	73							
Q2bii	85	15							
Q3ai	14	38	48						
Q3aii	93	7							
Q3aiii	69	31							
Q3b	8	25	68						
Q3c	73	27							
Q4ai	59	41							
Q4aii	37	63							
Q4aiii	36	64							
Q4b	56	21	22						
Q4c	34	66							
Q5ai	46	23	31						
Q5aii	52	48							
Q5aiii	37	63							
Q5bi	34	66							
Q5bii	35	65							
Q5ci	39	61							
Q5cii	25	27	47						
Q6a	51	49							
Q6b	39	61							
Q6ci	44	56							
Q6cii	67	33							
Q7ai	48	52							
Q7aii	36	64							
Q7aiii	19	55	26						
Q7bi	50	50							
Q7bii	70	30							
Q8ai	41	59							
Q8aii	50	50							
Q8b	61	39							
Q8ci	54	46							
Q8cii	48	52							
Q8di	70	30							
Q8dii	79	21							
Q9a	63	26	11						
Q9bi	67	33							
Q9bii	50	50							
Q9biii	49	51							
Q9c	61	39							
Q10ai	43	57							
Q10aii	81	19							
Q10aiii	52	48							
Q10aiv	57	43							
Q10bi	42	35	23						
Q10bii	19	81							
Q11a	33	67							
Q11b	74	26							
Q11c	36	64							
Q11di	51	49							
Q11dii	78	22							
Q11e	53	47							
Q12a	65	35							
Q12bi	26	74							
Q12bii	89	11							
Q12ci	22	78							
Q12cii	53	47							
Q12di	89	11							
Q12dii	63	28	10						
Q12diii	47	53							
Q13a	49	23	22	6					
Q13b	39	61							
Q14a	73	27							
Q14b	65	35							
Q14ci	15	85							
Q14cii	42	58							
Q14d	58	29	13						
Q15A	17	18	17	14	11	8	5	3	5
Q15B	9	8	9	13	16	14	11	10	10

2024 Higher Human Biology Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.8	81.8%	26
2	4	2.3	58.3%	39
3a	1	0.4	41.2%	28
3b	1	1	95.1%	13
3c	1	0.4	41.4%	31
3d	1	0.6	59.3%	20
3e	1	0.4	37.8%	35
4a	1	1	97.2%	16
4b	1	0.8	81.6%	24
4c	1	0.9	85.5%	28
4d	1	0.7	71.7%	30
5	1	0.3	27.9%	41
6	1	0.4	40.8%	32
7	3	1.1	36.0%	49
8	1	0.7	74.8%	17
Paper	20			

Item	Max Mark	0	1	2	3	4
1	1	18	82			
2	4	12	18	22	22	27
3a	1	59	41			
3b	1	5	95			
3c	1	59	41			
3d	1	41	59			
3e	1	62	38			
4a	1	3	97			
4b	1	18	82			
4c	1	14	86			
4d	1	28	72			
5	1	72	28			
6	1	59	41			
7	3	36	31	22	11	
8	1	25	75			
Paper	20					

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation
2	4	Underlying biology
3a	1	A brief summary of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data, including mean/average values, presented in a correctly produced table(s)
3d	1	Data relevant to the aim from an internet/literature source
3e	1	A citation and reference for a source of the internet/literature data
4a	1	An appropriate format from the options of bar graph or line graph.
4b	1	The axes of the graph have suitable scales.
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and a line (line graph) or clear bar tops (bar graph)
5	1	A valid comparison or an appropriate calculation linked to the aim of the investigation.
6	1	A valid conclusion that relates to the aim and is supported by all the data in the report.
7	3	Evaluation of the investigation
8	1	A clear and concise report, with an informative title.